

2Market Technical Report

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Word count 1,100 (excluding titles and appendices)

Introduction/ Problem Statement

2Market is a global supermarket, operating both online and instore. Using the 'IDEAL' problem-solving framework, their business problem/ opportunity can be **identified** as being that they do not currently understand their customer purchasing behaviour, and therefore, are potentially engaging in inefficient and ineffective advertising efforts, which could be preventing them from maximising sales revenue. Their goals can be **defined** as being to understand what their customer demographics are, which advertising channels are most effective, and which products are bestsellers per demographic. The solution being **explored** is the analysis of internal data on customers, advertising, and product sales to identify insights. However, the analysis is limited because neither additional internal data nor external data on the market/ industry/ sector is provided for consideration.

Analytical Approach

For the purposes of this analysis, it is assumed that the current year is 2015, because the date values within the 'Dt_Customer' column are between 2012 and 2014. It is also assumed that a customer can be a maximum of 120 years old, meaning the lowest acceptable cell value within the 'Year_Birth' column would be "1895". It is also assumed that a single customer may have registered with 2Market more than once and in more than one location globally, meaning they may have duplicate records within the dataset. For this reason, the 'ID' column and 'Country' column have been excluded when searching for duplicated rows within the dataset.

Microsoft Excel has been used for the initial cleaning and preparation of the data because there are only 2,216 records within the dataset, meaning Excel can handle this activity efficiently.

The following Excel functions and commands were used to clean and prepare the data:

Characteristic	Function/ Command	Target
Accuracy	1. Spellcheck 2. Find and replace 3. Remove blank spaces using =TRIM 4. Replace incorrect cell values with #N/A	1. All cells 2. Cells containing "\$" 3. All cells 4. 'Year_Birth' column cells containing years "<1895"
Completeness	1. Find all blank cells using 'go to; special'	1. All cells
Consistency	1. Convert all numbers stored as text into numbers using the 'text to columns' command	1. All cells containing numbers, currency, and dates, except IDs
Uniqueness	1. Highlight errors using the 'conditional formatting; new rule; only cells that contain errors' command 2. Highlight and remove all fully duplicated rows using the 'conditional formatting; highlight cell rules; duplicate values' command and the 'remove duplicates' command 3. Remove all conditional formatting using the 'conditional formatting; clear rules' command	1. All cells 2. All columns except 'ID' and 'Country' 3. All cells
Timeliness	1. Check date formats using the 'format cells; date' command 2. Check date relevance of cells	1. 'Dt_Customer' column cells 2. 'Recency' column cells

See [‘Appendix 1’](#) for additional steps taken to clean and prepare the data in Excel.

This cleaning and preparation resulted in the '\$' symbol being removed from 2,216 cells, all blank spaces being removed from all cells, the birth year of one customer being changed from "1894" to "#N/A" (and their age being changed from "121" to "#N/A"), all numbers (except IDs) being converted from text into numbers, and 182 fully duplicated records being removed. No spelling errors or blank cells were found, and no 'Dt_Customer' column cell values needed to be converted into dates. It is possible that the 'Recency' column cell values may have previously been calculated with reference to date columns, but this cannot be verified from the dataset.

See [‘Appendix 2’](#) for initial exploratory analysis undertaken in Excel and the related charts.

The cleaned marketing_data and the original ad_data csv files were then imported to Tableau and joined with a left join, so the only rows that were retained were those which contained common customer IDs in both datasets. The six product columns within the marketing_data were then pivoted to become one combined 'Product Category' column with an associated 'Sales Revenue' column. See [‘Appendix 3’](#) for details of the calculated fields created to support data exploration and visualisation.

Note that SQL was not used to clean, prepare, or join data prior to it being imported to Tableau. However, additional data manipulation and exploratory analysis was undertaken in SQL after work in Tableau had already begun. See [‘Appendix 4’](#) for full details.

Dashboard Design

Three dashboards were created for 2Market; one focusing on customer demographics, one on product categories, and one on customer purchasing behaviour by age range. This was so enough relevant visualisations could be shown to sufficiently answer each of the three research questions.

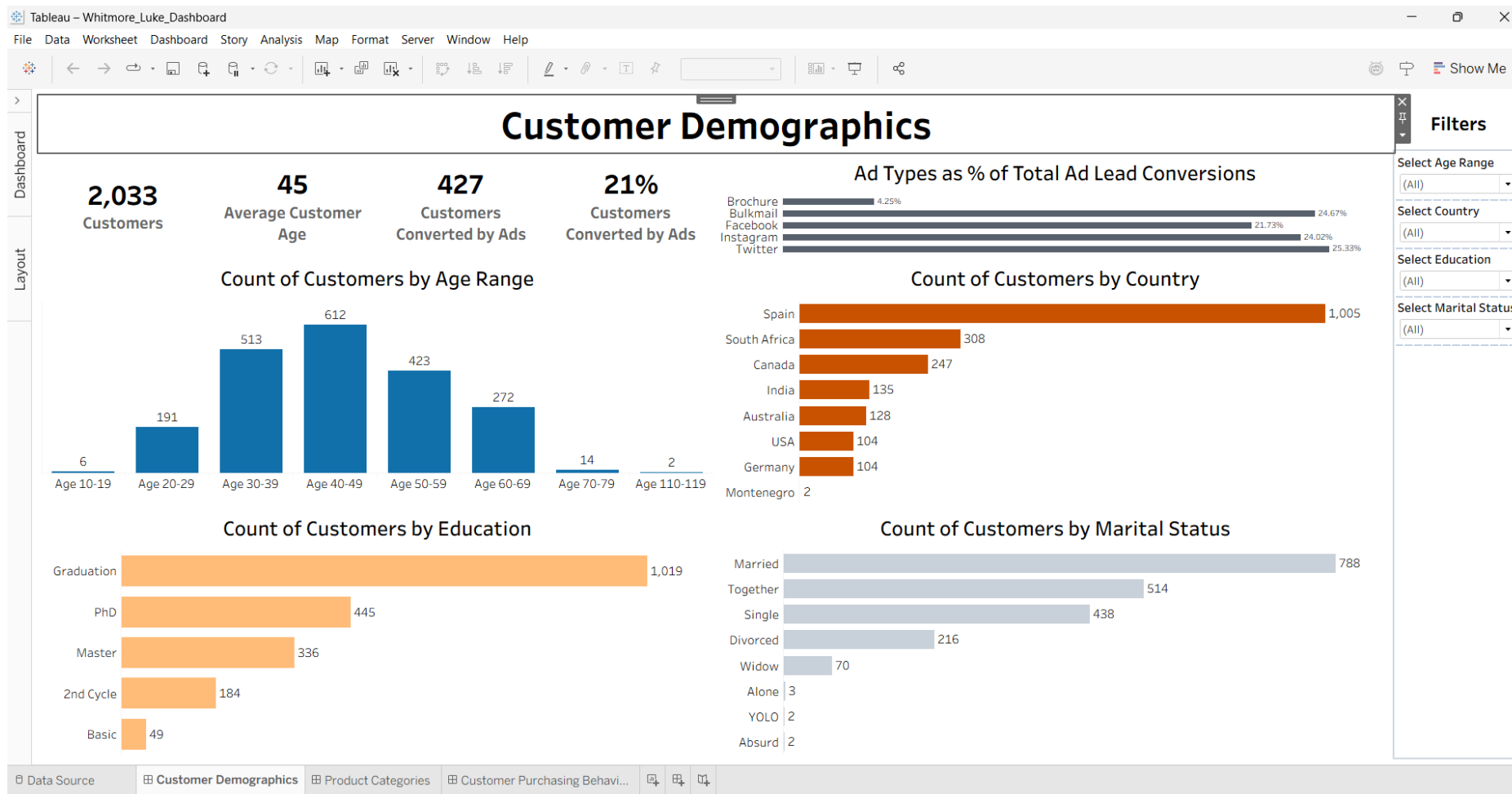
Cards were included to show high-level data that could quickly be viewed by any interested stakeholder, e.g., an executive director. Four to five bar charts were included in each dashboard. Bar charts were chosen because they were seen as the easiest way to show comparisons between data points (customer segments, sales, etc.)

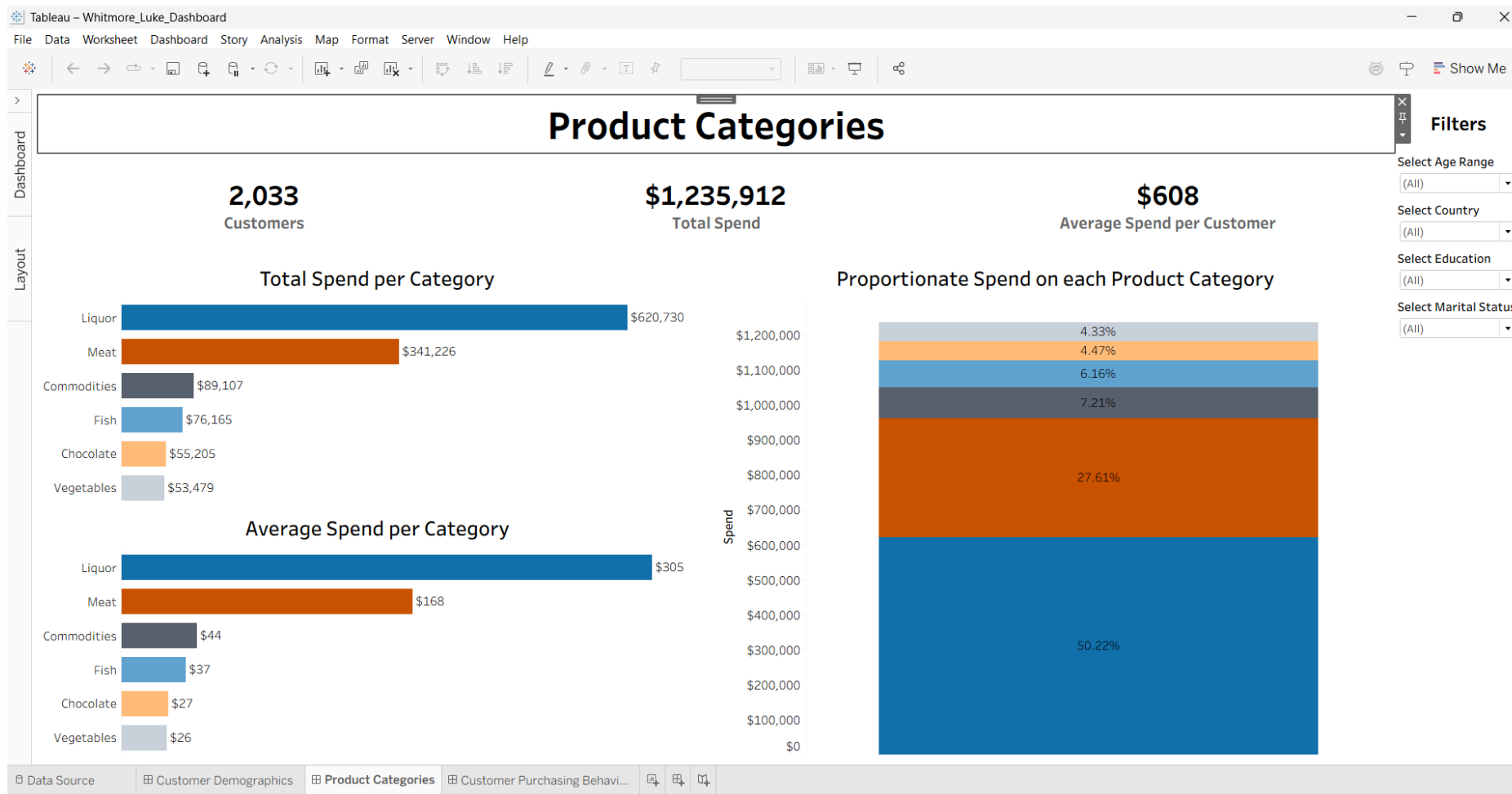
Filters were included to enable users, e.g., marketing managers, to drill down by customer segment and/or product category. The filter titles include the word “Select” so the user would know this is an action they could take.

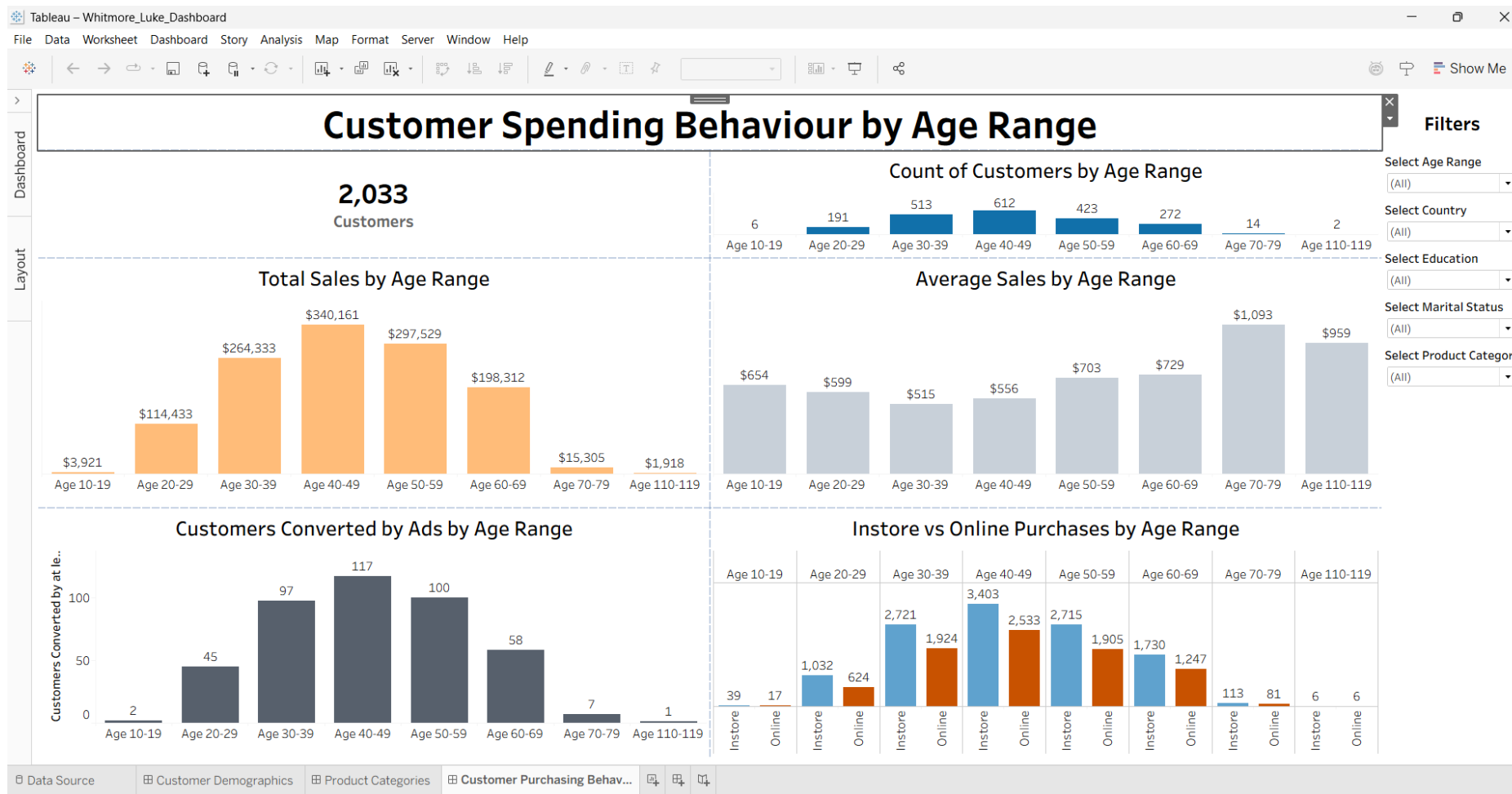
Font colour was set to default (black), and the colour-blind colour palette was applied to all visualisations, to improve accessibility.

Data labels and tooltips were added to several visualisations to provide additional relevant details that may be of interest to the user, e.g., % of total customers.

No captions were included because it was felt that the visualisation titles were explanatory enough to warrant omitting these.







Insight: Largest Customer Segments

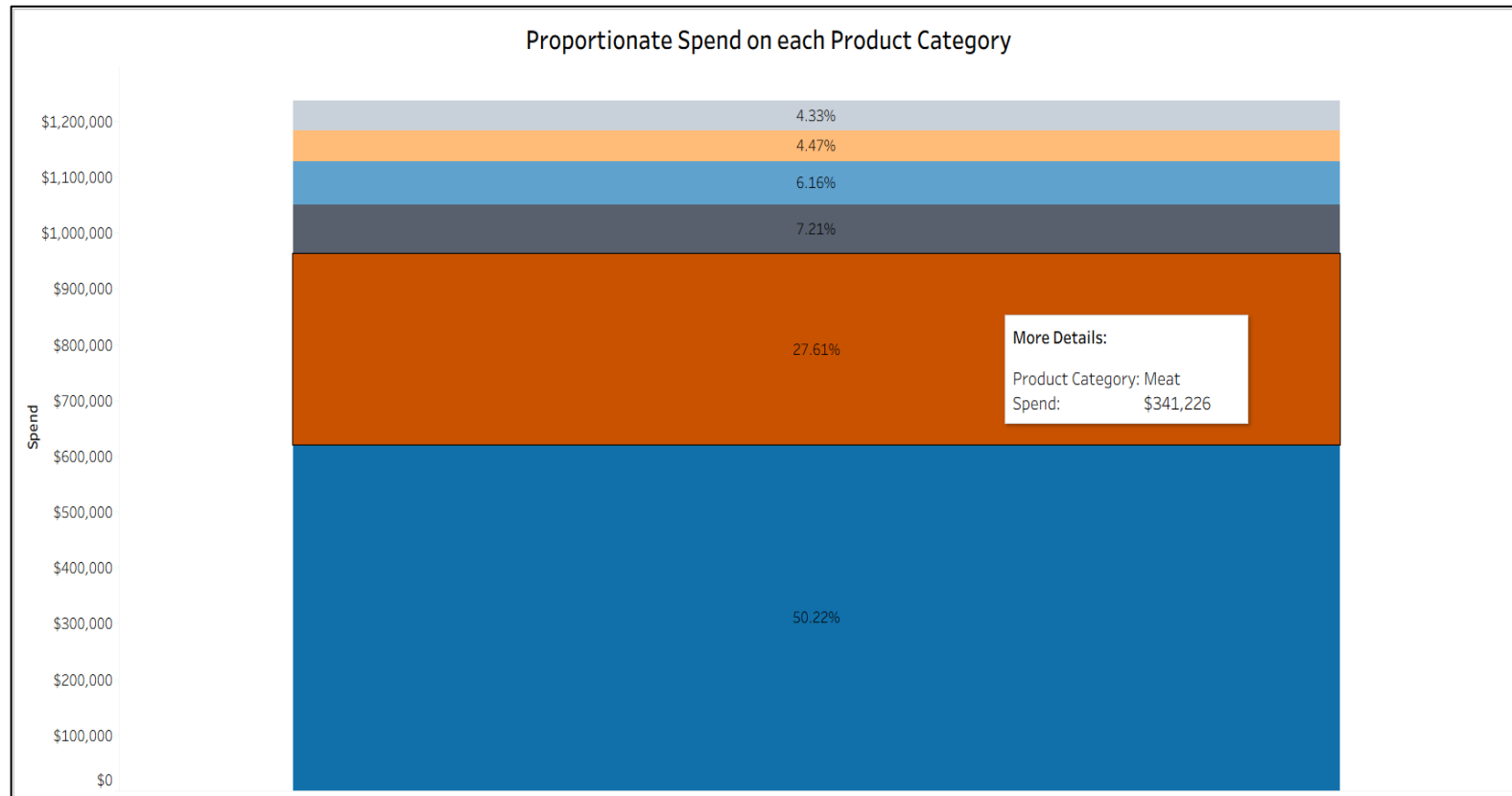
The five largest customer segments, which also account for the five highest total spends, are detailed below:

Customer Segment Size Ranking (out of 21 Segments)	Customer Segment	Number of Customers	% of Total Customers	Spend	% of Total Spend	Customer Segment Spend Ranking (out of 21 Segments)	Most Effective Ad Type
1	Education: Graduation	1011	51.74%	\$628,526	52.18%	1	Instagram
2	Country: Spain	966	49.44%	\$592,448	49.18%	2	Twitter
3	Marital Status: Married	762	39.00%	\$456,740	37.92%	3	Twitter
4	Age Range: 40-49	600	30.71%	\$338,759	28.12%	4	Twitter
5	Marital Status: Together	501	25.64%	\$308,986	25.65%	5	Twitter

This is useful for 2Market because knowing who most of their customers are and where most of their sales revenue comes from, as well as how much they rely on the business of any one segment, can help inform decisions about which customer segments to try to retain and grow.

Insight: Bestselling Products

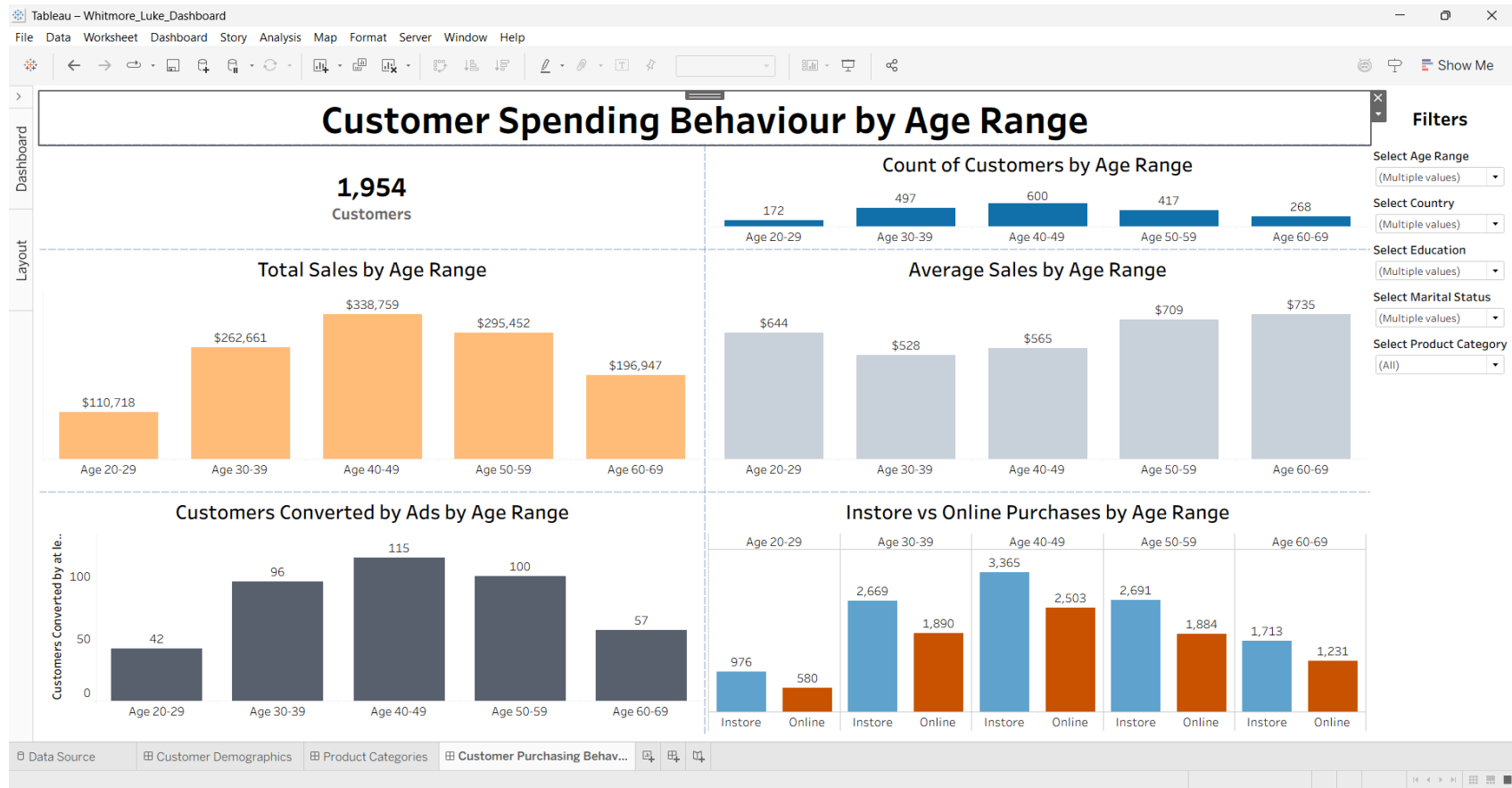
Liquor accounts for 50% of sales revenue and meat accounts for 28%, meaning more than three quarters of 2Market's sales are tied to these products. Furthermore, every customer segment spends most of their money on liquor, followed by meat.

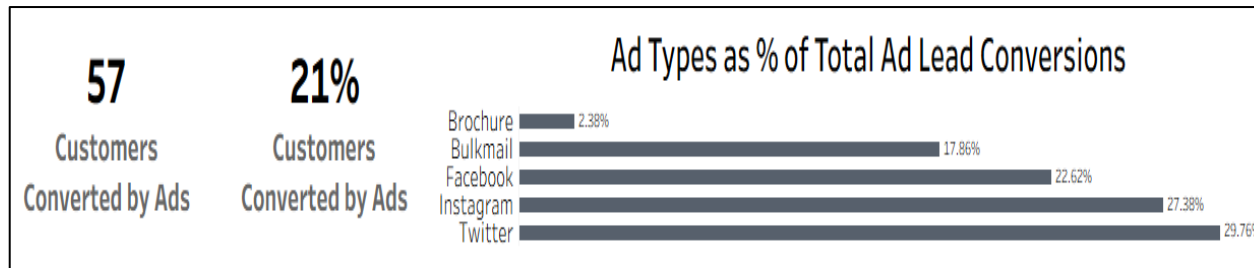


This is useful for 2Market to know because knowing which products generate the most sales revenue can help inform decisions, for example, which products to advertise promotional offers on, or how to spread the risk of being unable to sell any one product by trying to increase sales of others.

Insight: Age Ranges with Greatest Opportunity

Customers age 40-49 generated the most sales revenue in total. However, customers age 60-69 spent the most on average per customer. Additional details of customers age 60-69 are included below:





By filtering the data by product category, we can also see that, on average, customers age 60-69 spent the most on commodities, fish, and liquor out of all age ranges, whilst customers age 20-29 spent the most on chocolate, meat, and vegetables out of all age ranges.

Product Category	Highest Average Spender	Average Spend per Customer	Customer's Preferred Means of Shopping	Most Effective Ad Type
Chocolate	Age 20-29	\$31	Instore (63%)	Instagram
Commodities	Age 60-69	\$51	Instore (58%)	Twitter
Fish	Age 60-69	\$42	Instore (58%)	Twitter
Liquor	Age 60-69	\$390	Instore (58%)	Twitter
Meat	Age 20-29	\$218	Instore (63%)	Instagram
Vegetables	Age 20-29	\$32	Instore (63%)	Instagram

Knowing this could help 2Market to target their advertising of specific products to specific customer segments via specific advertising channels, to increase their customers base, and therefore, their sales revenue.

Recommendations and Further Actions

To enhance this analysis, it is recommended that 2Market provides additional data on all sales within the past year, so time series charts can be generated to identify more meaningful trends. The current data on the most recent sale of each customer does not account for natural attrition (due to relocation, death, change in shopping habits, etc.), so its relevance to the present day is limited (because some sales took place up to two years ago).

It is also recommended that 2Market consolidates duplicate customer records within their database. 182 were found during this analysis, meaning to rely on the data in its current state could lead to impaired future decision-making.

Appendices

1. Additional steps taken to clean and prepare the data in Excel:

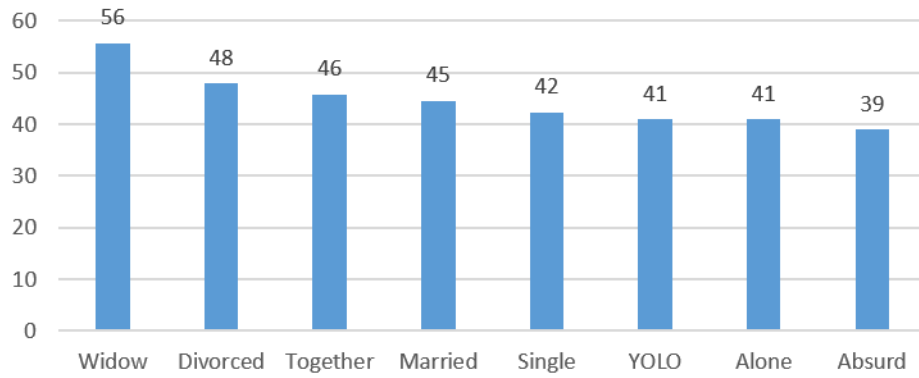
- a. Insert a new column called “Customer_Age” between the “Year_Birth” and “Education” columns, then within the first cell of the new column, insert the formula =IFERROR(SUM(2015-B2),0), and then copy this to all cells in the column to include customer ages within all records
- b. Insert a new column called “Income” between the “Income” and “Kidhome” columns, then copy and paste all cell values from the original Income column into the new column
- c. Insert two new cells between the “Year_Birth” and “Customer_Age” columns, then cut and paste the two “Income” columns into the two new cells, and then delete the previous two newly empty columns
- d. Insert five new rows above the dataset, then copy and paste the column headings from the dataset into the first new row, leaving the other four new rows blank (for advanced filtering)

2. Initial exploratory analysis in Excel

Function/ Command	Target	Result
=AGGREGATE(1,7)	‘Customer_Age’ column cells	The average customer age is 45 years old
Pivot Table: • Move ‘Marital_Status’ to rows • Move ‘Customer_Age’ to values, then click ‘Value Field Settings’, then select ‘Average’	Sheet1!\$C\$3	The average customer age per marital status is: • Absurd = 39 years old • Alone = 41 years old • Divorced = 48 years old • Married = 45 years old • Single = 42 years old • Together = 46 years old • Widow = 56 years old • YOLO = 41 years old
Filter	‘Income’ column cells	The average customer age of those customers within the income band \$90k - \$100k is 43 years old
Advanced Filtering	‘Income’ column cells	The income band with the highest average customer age is \$50k - \$60k, 49 years old, and the income band with the lowest average customer age is \$10k - \$20k, 37 years old
Scatterplot Chart	‘Customer_Age’ and ‘Income’ column cells	There is no correlation between customer age and income

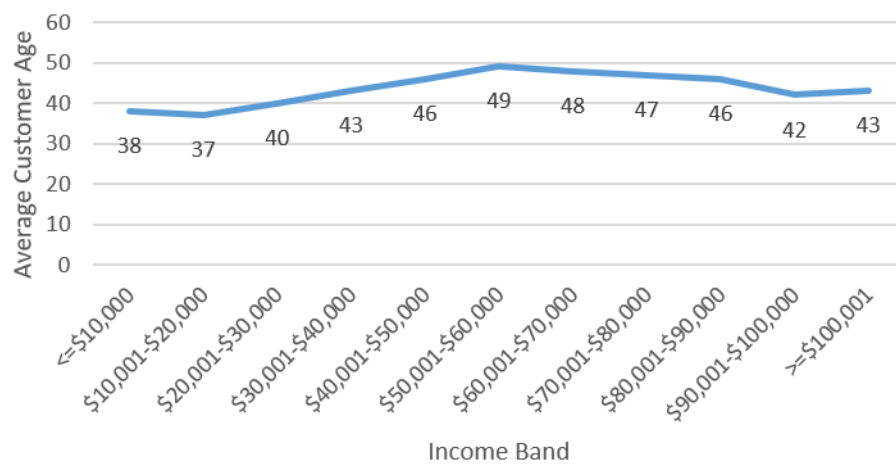
Average of Customer_Age

Average Customer Age by Marital Status

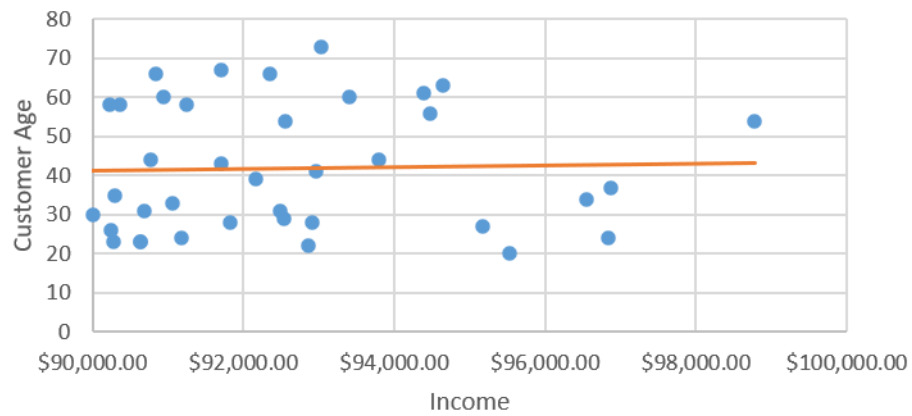


Marital_Status ▾

Average Customer Age by Income Band



Relationship between Customer Age and Income
between \$90k and \$100k



3. Calculated Fields created in Tableau

Field Name	Calculation	Rationale
Customer Age	2015 – [Year_birth]	2Market needs to know how old each of its customers is
Customers Converted by at least one Ad	{ FIXED [ID] : MAX([Total Customers Converted by an Ad]) }	2Market needs to know how many customers were successfully converted by an ad
% of Customers Converted by at least one Ad	SUM([Customers Converted by at least one Ad]) / COUNTD([ID])	2Market needs to know the % of distinct customers that were successfully converted by an ad
Brochure Conv.	COUNTD(IF [Brochure ad] = 1 THEN [ID] END)	Each customer ID has been duplicated, once for each product category, but 2Market needs to know how many distinct customers have been successfully converted by each type of ad
Bulkmail Conv.	COUNTD(IF [Bulkmail ad] = 1 THEN [ID] END)	
Facebook Conv.	COUNTD(IF [Facebook ad] = 1 THEN [ID] END)	
Instagram Conv.	COUNTD(IF [Instagram ad] = 1 THEN [ID] END)	
Twitter Conv.	COUNTD(IF [Twitter ad] = 1 THEN [ID] END)	
Total Conversions	[Brochure Conv.]+[Bulkmail Conv.]+[Facebook Conv.]+[Instagram Conv.]+[Twitter Conv.]	2Market needs to know the total number of distinct customers that have been successfully converted by all ads
% Converted via Brochure	[Brochure Conv.]/COUNTD([ID])	2Market needs to know the % of all distinct customers that have been successfully converted by each type of ad
% Converted via Bulkmail	[Bulkmail Conv.]/COUNTD([ID])	
% Converted via Facebook	[Facebook Conv.]/COUNTD([ID])	
% Converted via Instagram	[Instagram Conv.]/COUNTD([ID])	
% Converted via Twitter	[Twitter Conv.]/COUNTD([ID])	
Sales per Customer	{ FIXED [ID] : SUM([Sales Revenue]) }	Each customer ID has been duplicated, once for each product category, but 2Market needs to know the average total spend per customer
Instore Purchases	SUM([Num Walkin Pur])/6	2Market needs to know the number of purchases by distinct customers that were made online vs. instore
Online Purchases	SUM([Num Web Buy])/6	

4. Additional data manipulation and exploratory analysis in SQL:

SQL was used to create a database called “2Market” and the following syntax was written to create two tables, one for Marketing Data and one for Ad Data:

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x 2Market/postgres@PostgreSQL 18* x 2Market/postgres... x

2Market/postgres@PostgreSQL 18

No limit

Query Query History

```
1 CREATE TABLE "Marketing Data"(  
2 ID BIGSERIAL PRIMARY KEY,  
3 Year_Birth INTEGER,  
4 Education VARCHAR(20),  
5 Marital_Status VARCHAR(20),  
6 Income INTEGER,  
7 Kidhome INTEGER,  
8 Teenhome INTEGER,  
9 Dt_Customer DATE,  
10 Recency INTEGER,  
11 AmtLiq INTEGER,  
12 AmtVeg INTEGER,  
13 AmtNonVeg INTEGER,  
14 AmtPes INTEGER,  
15 AmtChocolates INTEGER,  
16 AmtComm INTEGER,  
17 NumDeals INTEGER,  
18 NumWebBuy INTEGER,  
19 NumWalkinPur INTEGER,  
20 NumVisits INTEGER,  
21 Response INTEGER,  
22 Complain INTEGER,  
23 Country VARCHAR(5),  
24 Count_success INTEGER);
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 259 msec.

Servers > PostgreSQL 18 > Databases > 2Market > Schemas > public > Tables > Ad Data

CRLF Ln 24, Col 24

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x 2Market/postgres... x 2Market/postgres@PostgreSQL 18* x

2Market/postgres@PostgreSQL 18

No limit

Query Query History

```
1 CREATE TABLE "Ad Data"(  
2 ID BIGSERIAL PRIMARY KEY,  
3 Bulkmail_ad INTEGER,  
4 Twitter_ad INTEGER,  
5 Instagram_ad INTEGER,  
6 Facebook_ad INTEGER,  
7 Brochure_ad INTEGER);
```

Data Output Messages Notifications

CREATE TABLE

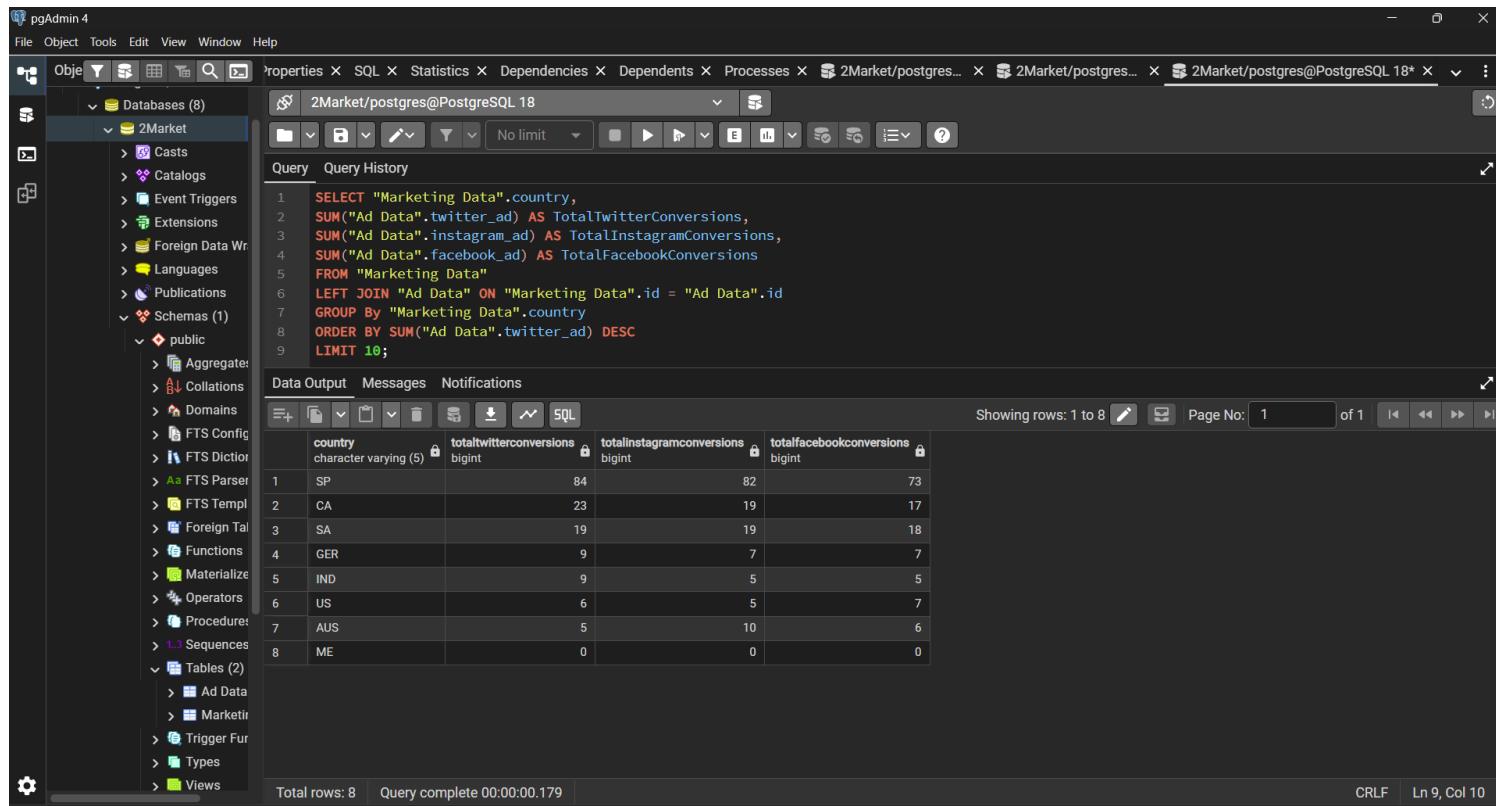
Query returned successfully in 246 msec.

Servers > PostgreSQL 18 > Databases > 2Market > Schemas > public > Tables > Ad Data

CRLF Ln 7, Col 22

The cleaned marketing_data and the original ad_data was then imported to each of the tables and joined using a left join to ensure that the only rows within the table were those that had been retained following the earlier cleaning and preparation of the marketing_data. The table was then queried to answer the following questions:

- Which social media platform (Twitter, Instagram, or Facebook) is the most effective method of advertising in each country?
- Which social media platform is the most effective method of advertising based on marital status?
- Which social media platform(s) seem(s) to be the most effective per country?



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including Databases (8), 2Market, and public. The main pane shows a SQL query executed against the 2Market/postgres@PostgreSQL 18 database. The query is as follows:

```

1 SELECT "Marketing Data".country,
2 SUM("Ad Data".twitter_ad) AS TotalTwitterConversions,
3 SUM("Ad Data".instagram_ad) AS TotalInstagramConversions,
4 SUM("Ad Data".facebook_ad) AS TotalFacebookConversions
5 FROM "Marketing Data"
6 LEFT JOIN "Ad Data" ON "Marketing Data".id = "Ad Data".id
7 GROUP BY "Marketing Data".country
8 ORDER BY SUM("Ad Data".twitter_ad) DESC
9 LIMIT 10;

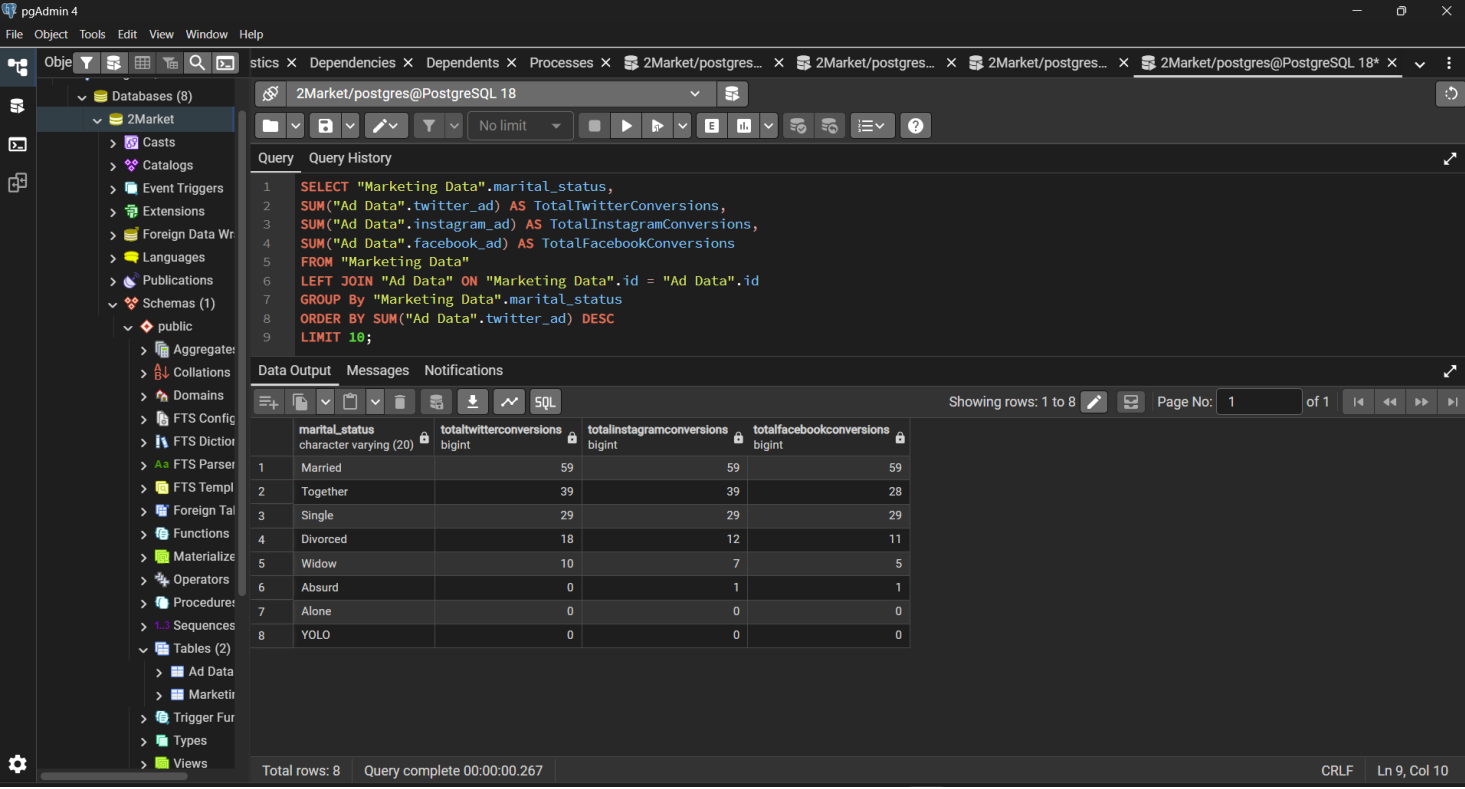
```

The results are displayed in a table with the following columns: country, totaltwitterconversions, totalinstagramconversions, and totalfacebookconversions. The table shows 8 rows of data, ordered by totaltwitterconversions in descending order.

	country	totaltwitterconversions	totalinstagramconversions	totalfacebookconversions
1	SP	84	82	73
2	CA	23	19	17
3	SA	19	19	18
4	GER	9	7	7
5	IND	9	5	5
6	US	6	5	7
7	AUS	5	10	6
8	ME	0	0	0

The status bar at the bottom indicates "Total rows: 8" and "Query complete 00:00:00.179".

Twitter was the most effective form of social media advertising in all countries except the USA (Facebook was the most effective), Australia (Instagram was the most effective), and Montenegro (social media had no effect). Instagram was as effective as Twitter in South Africa.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is as follows:

```

1 SELECT "Marketing Data".marital_status,
2 SUM("Ad Data".twitter_ad) AS TotalTwitterConversions,
3 SUM("Ad Data".instagram_ad) AS TotalInstagramConversions,
4 SUM("Ad Data".facebook_ad) AS TotalFacebookConversions
5 FROM "Marketing Data"
6 LEFT JOIN "Ad Data" ON "Marketing Data".id = "Ad Data".id
7 GROUP BY "Marketing Data".marital_status
8 ORDER BY SUM("Ad Data".twitter_ad) DESC
9 LIMIT 10;

```

The results are displayed in a table with the following columns: marital_status, totaltwitterconversions, totalinstagramconversions, and totalfacebookconversions. The table shows 8 rows of data, ordered by totaltwitterconversions in descending order.

marital_status	totaltwitterconversions	totalinstagramconversions	totalfacebookconversions
Married	59	59	59
Together	39	39	28
Single	29	29	29
Divorced	18	12	11
Widow	10	7	5
Absurd	0	1	1
Alone	0	0	0
YOLO	0	0	0

Total rows: 8 Query complete 00:00:00.267

All social media ad types were equally effective for customers with the marital statuses “married”, “single”, “alone”, or “YOLO”. Twitter was the most effective form of social media advertising customers with the marital statuses “together”, “divorced” or “widowed”. Instagram was as effective as Twitter with customers with the marital status “together”.

pgAdmin 4

File Object Tools Edit View Window Help

2Market/postgres@PostgreSQL 18

Query

```

1 SELECT "Marketing Data".country,
2 SUM("Marketing Data".amtliq + "Marketing Data".amtveg + "Marketing Data".amtnonveg + "Marketing Data".amtpep + "Marketing Data".amtchocolates +
3 "Marketing Data".amtcomm) AS TotalSales,
4 SUM("Ad Data".facebook_ad) AS FacebookAds,
5 SUM("Marketing Data".amtliq + "Marketing Data".amtveg + "Marketing Data".amtnonveg + "Marketing Data".amtpep + "Marketing Data".amtchocolates +
6 "Marketing Data".amtcomm) / NULLIF(SUM("Ad Data".facebook_ad), 0) AS SalesPerFacebookAd,
7 SUM("Ad Data".instagram_ad) AS InstagramAds,
8 SUM("Marketing Data".amtliq + "Marketing Data".amtveg + "Marketing Data".amtnonveg + "Marketing Data".amtpep + "Marketing Data".amtchocolates +
9 "Marketing Data".amtcomm) / NULLIF(SUM("Ad Data".instagram_ad), 0) AS SalesPerInstagramAd,
10 SUM("Ad Data".twitter_ad) AS TwitterAds,
11 SUM("Marketing Data".amtliq + "Marketing Data".amtveg + "Marketing Data".amtnonveg + "Marketing Data".amtpep + "Marketing Data".amtchocolates +
12 "Marketing Data".amtcomm) / NULLIF(SUM("Ad Data".twitter_ad), 0) AS SalesPerTwitterAd,
13 SUM("Ad Data".facebook_ad + "Ad Data".instagram_ad + "Ad Data".twitter_ad) AS TotalAds,
14 SUM("Marketing Data".amtliq + "Marketing Data".amtveg + "Marketing Data".amtnonveg + "Marketing Data".amtpep + "Marketing Data".amtchocolates +
15 "Marketing Data".amtcomm) / NULLIF(SUM("Ad Data".facebook_ad + "Ad Data".instagram_ad + "Ad Data".twitter_ad), 0) AS SalesPerAd
16 FROM "Marketing Data"
17 LEFT JOIN "Ad Data" ON "Marketing Data".id = "Ad Data".id
18 GROUP BY "Marketing Data".country
19 ORDER BY SalesPerAd DESC;

```

Data Output

Showing rows: 1 to 8 Page No: 1 of 1

	country	totalsales	facebookads	salesperfacebookad	instagramads	salesperinstagramad	twitterads	salespertwitterad	totalads	salesperad
	character varying (5)	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint	bigint
1	ME	2132	0	[null]	0	[null]	0	[null]	0	[null]
2	IND	70519	5	14103	5	14103	9	7835	19	3711
3	US	66290	7	9470	5	13258	6	11048	18	3682
4	SA	192977	18	10720	19	10156	19	10156	56	3446
5	AUS	71673	6	11945	10	7167	5	14334	21	3413
Total rows: 8		Query complete 00:00:00.238								

CRLF Ln 19, Col 26

Facebook ads generated the most sales revenue per ad in India, South Africa, Germany, Canada, and Spain. Instagram ads generated as much sales revenue per ad as Facebook ads did in India and Germany. Instagram ads generated the most sales revenue per ad in the USA. Twitter ads generated the most sales revenue per ad in Australia. Social media ads generated no sales revenue in Montenegro. India's social media ads appeared to be the most efficient, generating \$3,711 of sales revenue per ad, whilst Spain's social media appeared to be the least efficient (excluding Montenegro), generating \$2,553 of sales revenue per ad.